

An Integrated Framework for Retail Business and Consumer Intelligence

Submitted by

Name: Divya Rubini. S

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1. Introduction

Retail businesses generate large volumes of transactional data through customers, orders, order items, products, stores, staff and inventory activities. This project focuses on organizing, cleaning, analyzing and visualizing retail sales data through an end-to-end analytics workflow.

The project combines Excel for data cleaning and preparation, SQL for relational database management and querying, Python for exploratory analysis and customer segmentation, and Power BI for interactive business reporting. The workflow follows the project task sheet from raw data preparation through dashboard development.

2. Problem Statement

In a competitive retail market, businesses need to monitor sales performance, understand customer purchase behavior, identify valuable customer groups, track inventory levels and evaluate product, store and staff performance. The project develops an end-to-end analytical solution that transforms raw transactional data into structured information that can be queried, analyzed and presented through an interactive Power BI dashboard.

3. Objectives

- Clean and prepare retail data using Excel.
- Build and manage a relational retail database using SQL.
- Apply SQL joins, aggregations, filters, grouping and ranking techniques.
- Load SQL data into Python for exploratory analysis.
- Calculate Recency, Frequency and Monetary (RFM) customer features.
- Create customer segments based on purchasing behavior.
- Export the segmentation results back to SQL.
- Connect the SQL database to Power BI and establish table relationships.
- Develop KPI cards, charts, maps, slicers and interactive dashboard pages.

- Integrate the complete Excel → SQL → Python → SQL → Power BI workflow.

4. Dataset Description

The project task sheet defines a relational Retail Sales Data structure containing the following tables:

Table	Purpose	Key Information
Customers	Customer master data	Customer details, contact and location
Orders	Order-level transactions	Order, customer, staff and store references
order items	Order line details	Products, quantity and pricing information
Products	Product information	Product, category and brand details
brands / categories	Product metadata	Brand and category classification
Stocks	Inventory information	Stock availability across stores
Stores	Store information	Store and geographical details
Staffs	Staff information	Employees associated with handled orders

5. Methodology

- **Excel – Data Preparation:** The raw data was cleaned, validate, formatted, and prepared for analysis. Derived columns and CSV files were also created.
- **SQL – Database Management:** The cleaned data was imported into MySQL. Tables and relationships were created, followed by SQL queries for data analysis.

- **Python – Customer Analysis:** Python was used to load and explore the data, calculate **Recency, Frequency, and Monetary (RFM)** values, and segment customers based on their purchasing behavior.
- **SQL – Segmentation Storage:** The customer segmentation results generated using Python were stored back in the SQL database for further use.
- **Power BI – Dashboard Development:** The final data was connected to Power BI to create the data model, KPIs, DAX measures, charts, slicers, and an interactive dashboard.

6. Software and Technologies Used

- **Microsoft Excel:** Used for cleaning, validating, formatting, and preparing the data.
- **MySQL Workbench:** Used to create the database, manage tables and relationships, and perform SQL analysis.
- **Python:** Used for data processing, Exploratory Data Analysis (EDA), RFM analysis, and customer segmentation using Pandas and NumPy.
- **Power BI:** Used for data modelling, DAX calculations, KPI creation, visualizations, filters, and interactive dashboard development.

7. Phase 1 – Excel: Data Cleaning & Preparation

- The first phase prepares the retail dataset for reliable database loading and analysis. The following tasks follow the project task sheet.
- Column headers were standardized using a consistent naming convention, such as snake case.
 - Reviewed the headers in each worksheet.
 - Removed inconsistent spacing and naming variations.
 - Applied a common naming pattern across related sheets.
- Duplicate records were checked and removed where necessary to avoid repeated transactions or master records.
 - Used Excel duplicate checking/removal tools.
 - Verified the remaining records after duplicate removal.

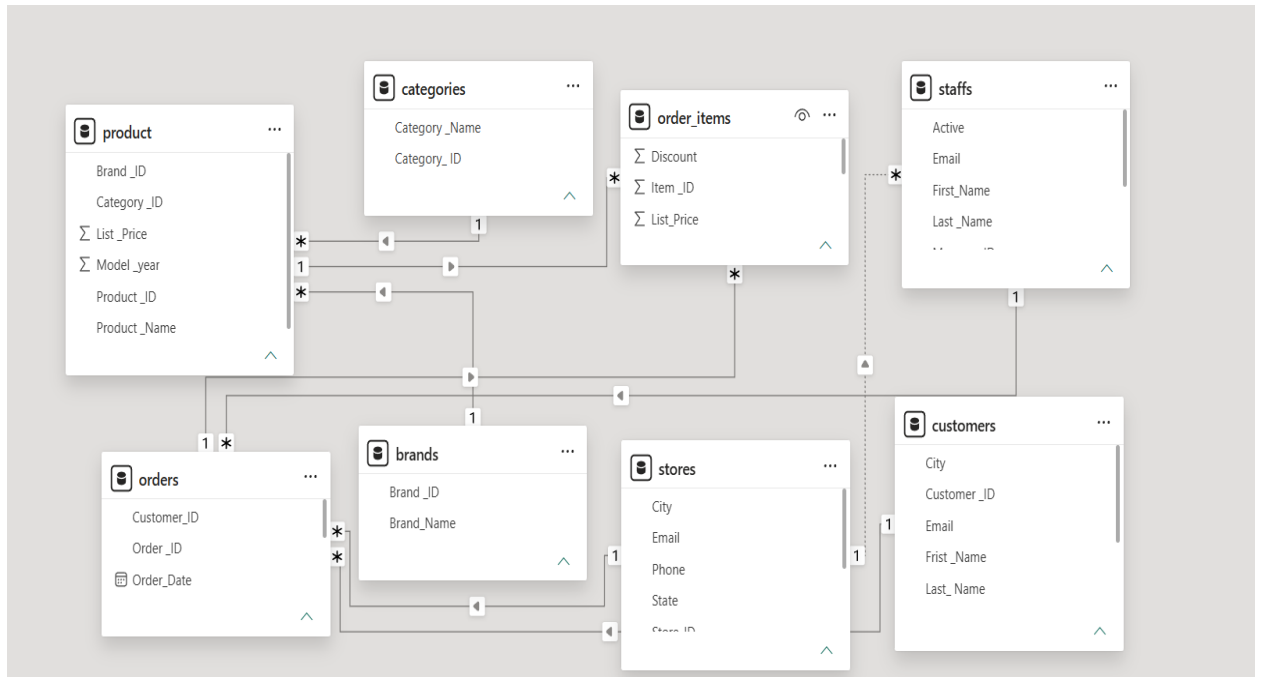
- Missing values were identified using filtering and replaced or retained according to the field requirement.
 - Used NA or an appropriate placeholder for fields requiring a non-blank value.
- Data Validation dropdowns were created to keep order status values consistent.
 - Configured a list containing Pending, Shipped and Delivered.
- A derived total price column was created to support sales calculations.
 - = (list price * quantity) – discount
- Product information was brought into the order item data using the VLOOKUP/XLOOKUP.
- A PivotTable was prepared to summarize sales across product categories.
- Pricing data was sorted and filtered to identify unusually high or low values for review.
- Cleaned worksheets were prepared as CSV files using table-specific filenames.

8. Phase 2 – SQL: Database Management and Querying

The SQL phase converts the cleaned files into a relational database and applies analytical queries to answer the requirements defined in the task sheet.

- The cleaned CSV files were imported into the corresponding MySQL tables.
- Relational tables were created in MySQL Workbench with primary keys, foreign keys and suitable data types.
 - Defined **primary keys**.
 - Defined **foreign key** relationships.
- **INNER JOIN** was used through common keys to combine order, item and product information.
- Products were ranked by total quantity sold using SUM, ORDER BY and LIMIT.
- A **CASE** expression was used to create spending categories based on total customer spend. A destination table named customer segments was prepared for the Python-

generated RFM segment labels.



9.Phase 3 – Python and Customer Segmentation

Python was used to load the SQL data, perform exploratory analysis, calculate RFM features and create customer segments. The project task sheet identifies RFM as Recency, Frequency and Monetary value.

- Use `df.describe()`, `df.info()`, `df.value_counts()` and basic charts to summarize the dataset.
- Exploratory Data Analysis was performed to understand data types, summary statistics, categorical frequencies and basic patterns.
- RFM metrics were calculated at customer level. Recency represents days since the customer's latest order, Frequency represents the number of orders, and Monetary represents the total purchase value.

Customer Segmentation

For the current dataset, the observed RFM ranges were

- Recency 1–1093 days
- Frequency 1–3 orders
- Monetary 109.99–24,488.90.
- The completed RFM Data Frame, including the Segment column, was exported from Python into MySQL as customer segments.

1.Created a SQL Alchemy connection to MySQL.

2. Confirmed that rfm_data contained Customer, Recency, Frequency, Monetary and Segment.
3. Verified the table in MySQL Workbench using SELECT and SHOW TABLES.

10. Phase 4 – Power BI: Visualization & Dashboarding

Power BI was used to connect to the SQL database, create relationships, build analytical visuals and provide interactive reporting. The following tasks follow the mandatory Power BI section of the task sheet.

The Power BI solution was developed using multiple report pages, with each page concentrating on a particular business analysis area.

The dashboard contains:

1. Home / Consolidated Dashboard
2. Customer Analysis & Purchasing Behavior
3. Product & Inventory Analysis
- 4 Sales Analysis

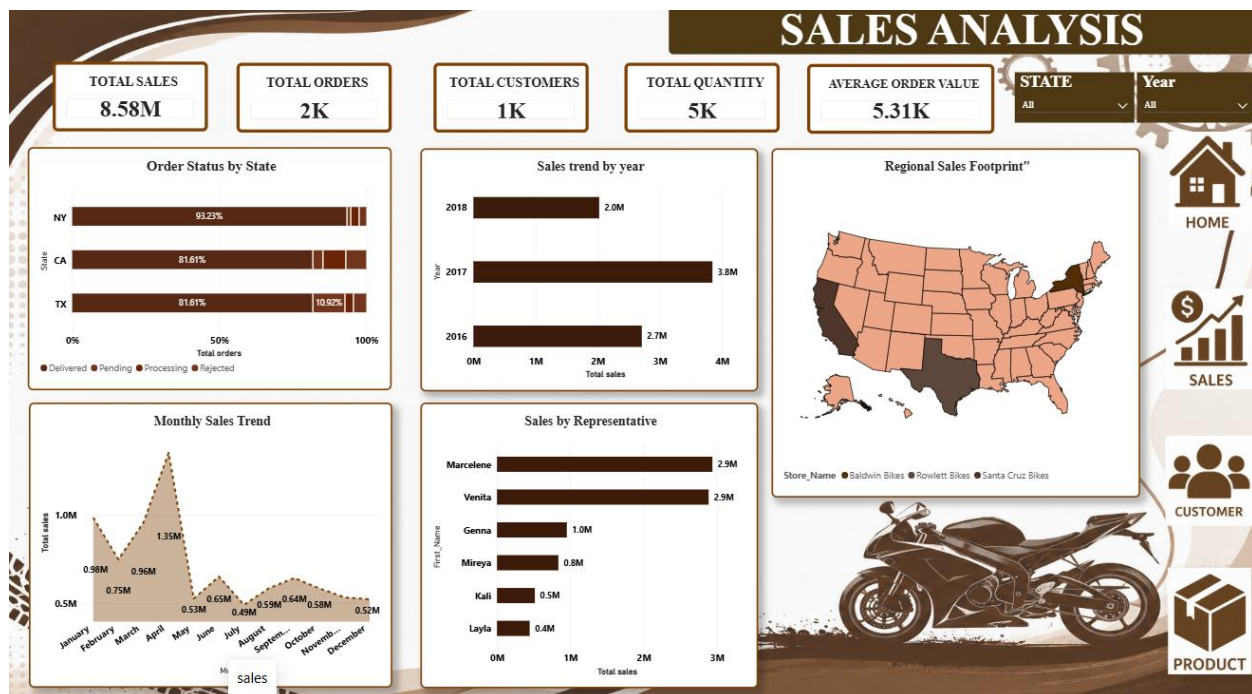
Each page contains appropriate visual elements such as KPI cards, charts, slicers, filters, maps, and navigation buttons. These features allow users to interact with the report and analyse retail sales information from different perspectives.

Dashboard Functionality

The Power BI dashboard incorporates several interactive features, including:

- **Dynamic KPI Cards** – Display important retail sales and customer metrics.
- **Slicers** – Allow users to filter information based on selected categories, stores, order status, and customer segments.
- **Navigation Buttons** – Allow users to move between different dashboard pages.
- **DAX Measures** – Perform customized calculations for sales, orders, customers, and other business metrics.
- **Interactive Charts and Maps** – Help analyse product, customer, store, regional, and sales patterns.
- **Consistent Formatting** – Maintains a professional and uniform dashboard appearance.

These functionalities make the dashboard interactive and allow users to explore retail sales performance, customer behavior, product performance, inventory, regional activity, staff performance, and customer segments efficiently.



11. Data Flow / End-to-End Pipeline

The completed workflow follows the sequence defined by the task sheet:

Excel Cleaning → SQL Database → Python EDA & RFM → SQL customer segments
→ Power BI

12. Project Deliverables

- Cleaned Excel/CSV files.
- SQL database creation and analytical query scripts.
- Python notebook/script containing EDA, RFM calculation and segmentation.
- customer segments table exported from Python to SQL.
- Power BI report with KPIs, charts, slicers and interactive analysis.
- Final documentation of the implemented workflow.

13. Conclusion

The Retail Sales Analytics project demonstrates an end-to-end approach to transforming raw retail transactional data into structured analytical information. Excel was used for data cleaning and preparation, SQL was used to create the relational database and perform analytical queries, Python was used for exploratory analysis and RFM-based customer segmentation, and Power BI was used to present the results through interactive dashboards.

The project also demonstrates the movement of analytical results between tools: cleaned data is loaded into SQL, customer RFM features are calculated in Python, segmentation results are exported back to SQL as customer segments, and the resulting table is incorporated into Power BI for segment-based reporting. This provides a structured and repeatable analytics pipeline covering data preparation, database management, analysis, segmentation and visualization.